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Varietal susceptibility overcomes climate change effects on the future trends of rice blast disease in Northern Italy

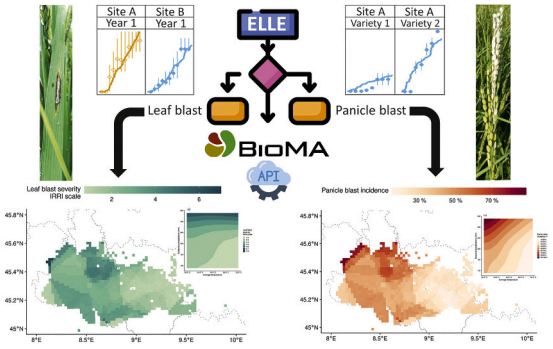
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HIGHLIGHTS

- The ELLE simulation model reproduces the dynamic of leaf and panicle blast severity and incidence during rice growing season.
- ELLE is calibrated/evaluated using data from multi- site and year surveys and spatialized observations at municipality level.
- Simulations under climate change scenarios highlighted key factors in modulating future trends of rice blast epidemics.
- Varietal susceptibility massively modulated rice blast epidemics, proving that breeding is essential against the disease.
- Effects of inter-annual weather variability outweigh mean changes, therefore in-season support is needed to limit rice blast.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Rice blast manifests with different symptoms according to the affected plant organ, i.e., leaf and panicle blast, the latter being the primary determinant of yield losses in Italy. Forecasting models have been developed to drive fungicides application and perform scenario analyses to plan legislative measures on plant protection. An assessment of the blast pathogen pressure evolution in the Italian rice area under climate change realizations has not been done yet.

OBJECTIVE: We developed the ELLE process-based forecasting model (ricE bLast simuLation modEl) to reproduce the dynamics of leaf and panicle blast severity and incidence to release a tool to perform in-season monitoring and scenario assessment. The forecasting model is delivered as a RESTful API and is ready to be used by third parties for operational monitoring activities. Here we present its application to assess future rice blast impacts in Italy after an extended calibration and evaluation with ground-truth data.

METHODS: The reference dataset for model testing comprised (a) a multi-year (2013–2015) and multi-site (three locations) experimental trial, where leaf and panicle blast were assessed on four Italian rice varieties with

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different levels of susceptibility to blast disease, and (b) historical disease data collected at the municipality level. We then performed spatially distributed simulations at 2 × 2 km resolution on the Northern Italian rice district, considering a short- (2030) and long- (2070) term time horizon and rice varieties representative of the current variability in phenology and susceptibility to blast disease within the Italian varietal landscape. The factorial combinations of climatic projections from four General Circulation Models and two extreme CO_2 Representative Concentration Pathways were used as model inputs in the scenario analysis.

RESULTS AND CONCLUSIONS: Model calibration and evaluation at the field level denoted an overall accurate reproduction of leaf blast severity (RRMSE <13%) and panicle blast incidence (RRMSE <17%). The climate change scenario analysis highlighted the varietal susceptibility as the critical factor determining leaf and panicle blast variability in the future, with no significant changes in 2030 and a slight decline in 2070.

SIGNIFICANCE: Besides delivering a reusable methodology to analyze the severity and incidence of leaf and panicle blast, our modelling study confirms that varietal susceptibility is the main feature to be pursued in rice breeding programs to counteract blast disease.

1. Introduction

Reliable forecasting models of plant diseases dynamics are essential to design optimized plant protection strategies (Savary et al., 2019) and support priority setting by policymakers in light of climatic changes (Donatelli et al., 2017). Available studies quantifying future dynamics of plant diseases are limited to few areas (Newbery et al., 2016), with wheat, grapevine, and oilseed rape as the most investigated crops (Juroszek and von Tiedemann, 2015). Scientists are still debating the effects of climate change on plant diseases, with significant implications on the development and survival of pathogens and host resistance to fungal infections (Elad and Pertot, 2014). The leading methodological approach found in the literature is simulation modelling, thanks to its capability to reproduce nonlinear processes underlying the pathogens' biological cycles in response to variable environmental conditions.

Rice pests and diseases are responsible for 10–30% of yield losses worldwide (Skamnioti and Gurr, 2009). Rice blast, caused by the fungus *Magnaporthe oryzae* B. Couch, is among the most widespread and destructive diseases in the major rice-growing areas (Kunova et al., 2013). Rice cultivation in Italy contributes to more than half of the European rice production (Organization, Agriculture and Food, 2012), is also seriously threatened by rice blast, forcing most farmers (over 75% of total rice area) to rely on fungicides (Di Tullio and Baldi, 2011). Rice blast is the major constraint of rice yield in Italy (Titone et al., 2015). Indeed, 14 of the top 20 most cultivated varieties, accounting for about 70% of the total rice-growing area, are moderate to highly susceptible to the rice blast pathogen (Titone et al., 2015; Mongiano et al., 2020; Ente Nazionale Risi, 2020). Two distinct symptomatologies are associated with rice blast according to the affected plant organs, i.e., the leaves and the panicles (Froyd et al., 1976). The pathogen can colonize the leaves before heading, causing necrotic lesions which reduce light absorption far beyond the symptomatic area (Bastiaans et al., 1994). During ripening, the fungus attacks the plant nodes and the panicle base (Gianessi and Williams, 2011), obstructing the remobilization of assimilates from the stems. This damage causes direct yield losses (Shim et al., 2005) and breakages to the rice grains, leading to decreased milling quality (Bregaglio et al., 2017).

Several research studies focused on climate change effects on leaf blast (e.g., Kim et al., 2015, Kim and Cho, 2016; Luo et al., 1995; Viswanath et al., 2017). Leaf blast severity, i.e., the percentage of host tissue affected by the disease, is indeed one of the main yield-reducing factors in most rice-producing countries; therefore, most modelling studies focused on this symptomatology (Katsantonis et al., 2017). On the contrary, panicle blast simulation usually targets disease incidence, i.e. the percentage of plants affected by the disease, and the consequent yield losses (Shim et al., 2005). Also, the combined consideration of leaf and panicle blast in a single simulation model has been pursued with varying complexity (Mousanejad et al., 2009; Bregaglio et al., 2016), the simplest represented by linear relationships between plant parameters and disease severity or incidence to estimate yield losses (Torres and Teng, 1993). The input datasets needed for the execution of current models generally comprise weather data (e.g., air temperature and relative humidity, Lanoiselet et al., 2002), as well as parameters modulating the epidemiological processes driving blast epidemics and plant physiological processes (e.g., leaf photosynthetic rate, van Van Ittersum et al., 2003). Surprisingly, information on rice varieties' susceptibility is often neglected in modelling studies, although genetic resistance is considered the most effective and economical method to control blast epidemics to date (Wu et al., 2015; Ning et al., 2020). Crop susceptibility has usually been assigned by default in blast models (e.g., LEAFBLAST, Choi et al., 1998; EPIBLAST, Kim and Kim, 1993). In contrast, recent studies performed with EPIRICE (Savary et al., 2012) assigned different categories of genetic resistance based on multi-year blast nursery tests in South Korea (Kim et al., 2015; Kim and Cho, 2016). The genetic component has also been considered in other plant disease model (Audsley et al., 2005). The level of varietal susceptibility to panicle blast in Italian rice varieties has been recently assessed (Titone et al., 2015; Mongiano et al., 2020), providing reference data to embed this information in modelling studies.

Despite the destructiveness of rice blast in rice-producing countries, to date, no modelling studies dealt with the impacts of climate change on the two symptomatologies. Releasing reliable figures on future rice blast disease trends in Italy under plausible climate change scenarios is essential to identify adaptation strategies and priorities in upcoming breeding programs. To this end, we present a climate change assessment performed with a process-based forecasting model (ELLE, RicE BLast SimuLation ModEl) of leaf blast severity and panicle blast incidence, including the effect of varietal susceptibility and phenological development. Model calibration and evaluation relied on multi-year and multi-site experimental trials, and on-field surveys carried out in the Italian primary rice-growing provinces. Spatially distributed simulations were executed over the whole Italian rice district under baseline climate and considering an ensemble of climate change projections in two time horizons centred in 2030 and 2070 to evaluate future rice blast trends in the target area. Our work provides quantitative figures to support policymakers in the law-making process for the Italian rice sector, while framing the context in which breeders will have to develop the upcoming rice breeding programs.

2. Materials and methods

2.1. Overview of the simulations study

The workflow of this modelling study is presented in Fig. 1.

The data used for the modelling tasks comprised 36 field datasets (four Italian rice varieties × three locations × three growing seasons), split in calibration and evaluation datasets via hold-out and bootstrap method (Section 2.3). Model results were evaluated by comparing the observations of leaf blast severity and panicle blast incidence, using standard evaluation metrics commonly employed in agroecological modelling studies (Section 2.3).

The climate change scenario analysis was performed via spatially

distributed simulations executed on the main Italian rice district using projections resulting from the combination of Four General Circulation Models (GCMs) and two extremes of the Representative Concentration Pathways (RCPs).

Future simulations were performed using a factorial combination of three blast susceptibility levels (Titone et al., 2015) and three crop cycle durations (Mongiano et al., 2019) in order to represent the actual variability found in current Italian rice cultivars (Section 2.4.2).

2.2. Modelling the seasonal dynamics of leaf and panicle blast disease

The ELLE model's flow diagram is shown in Fig. 2, presenting the simulated epidemiological processes, model parameters, and the input variables. ELLE was realized within the BioMA framework, extending the algorithms implemented in the *Diseases* components (Bregaglio and Donatelli, 2015).

Rather than being represented by the development of a new disease model, the novelty element consists of the coupling of existing sub-models, each targeting a specific epidemiological process to give a new modelling solution, which was then parametrized according to the biological requirements of the rice blast pathogen. The simulation of varietal susceptibility to rice blast and phenology dynamics was then implemented by specific approaches to modulate leaf and panicle blast dynamics according to host development (Sections 2.2 and 2.4.1).

The simulation of leaf blast severity begins on January 1, whereas panicle blast incidence starts to be simulated from rice flowering date, when panicles are exerted and start to be susceptible to the blast pathogen. In calibration, the starting date of panicle blast simulation was set according to phenological observations, whereas in climate change scenario analysis the rice flowering date was simulated according to Mongiano et al. (2019). A hydro-thermal threshold is used to estimate the onset of leaf blast symptoms (Bregaglio et al., 2016). Then, four sub-models reproduce the pathogen's main epidemiological processes (infection, latency, sporulation and dispersal). The estimation of leaf blast symptoms onset is performed with the accumulation of hydro-thermal time (IP_{state}, Eq. (1) and Eq. (2)) at an hourly time step (i, Eq. (1)).

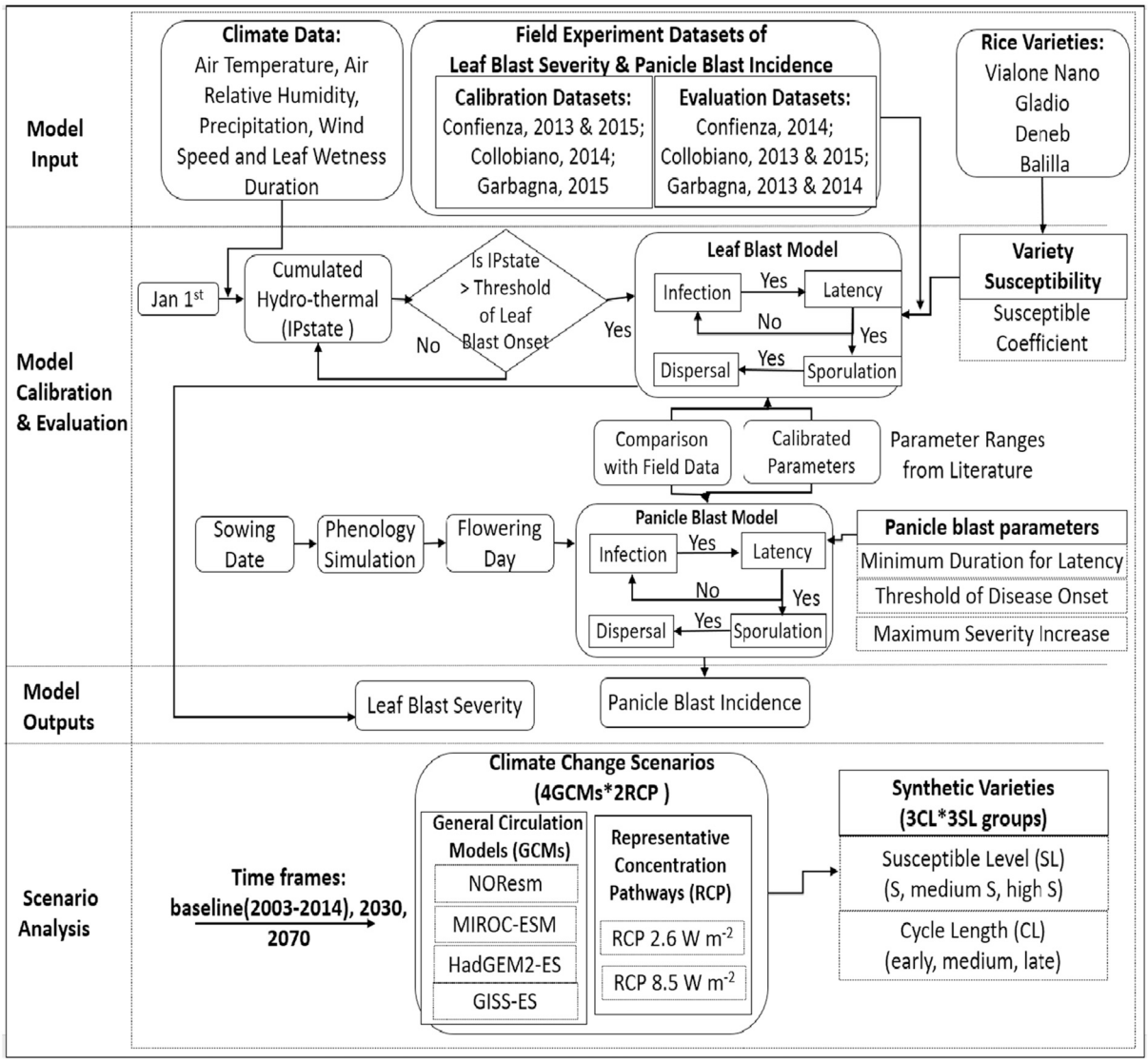


Fig. 1. Simulation experiment design: field datasets referred to four rice varieties with different blast susceptible levels grown in three sites in 2013–2015 growing seasons were split into calibration and evaluation subsets; the following epidemiological processes were simulated: infection, latency, sporulation and spore dispersal; simulated leaf blast severity and panicle blast incidence were compared to disease observations; the climate change scenarios comprised two time frames (2030, 2070), and were generated according to four general circulation models, considering two representative concentration pathways; future simulations were performed with synthetic rice varieties characterized by different blast susceptible level (high, moderate, low) and crop cycle length (early, medium, late).

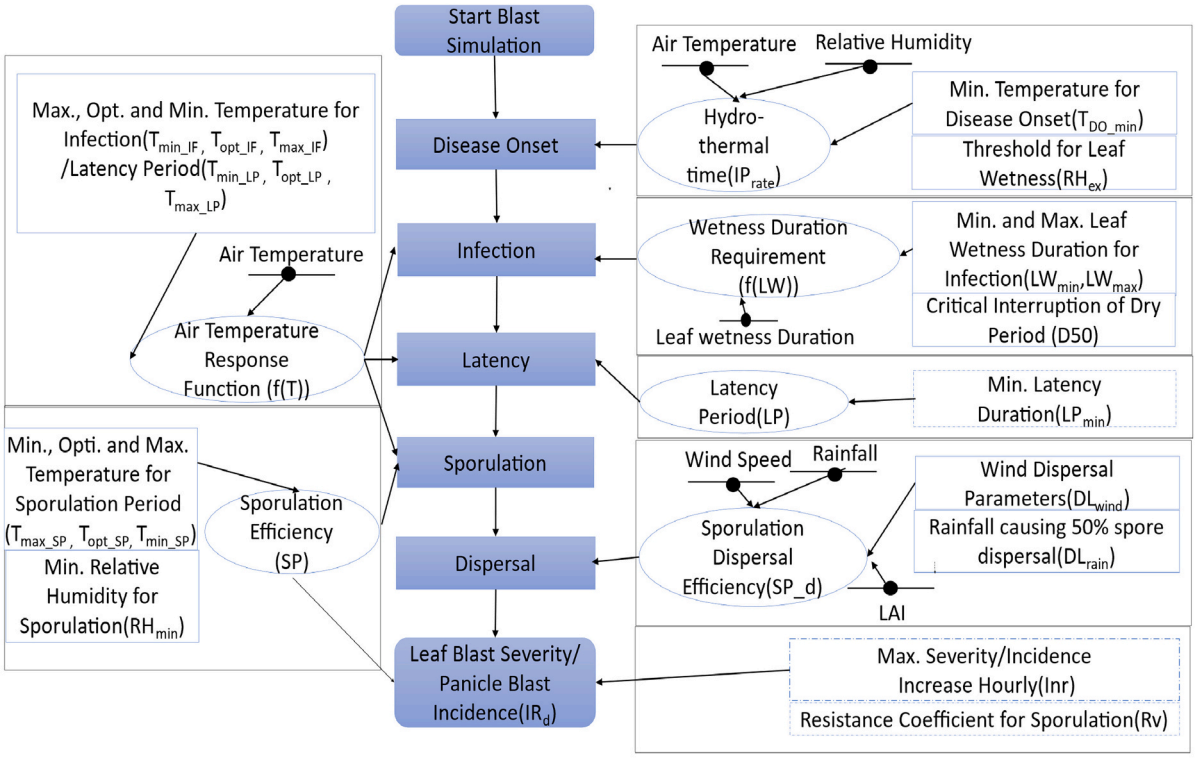


Fig. 2. Flow diagram of the ELLE model. Small crossed circles are input variables, blue boxes are epidemiological processes and model outputs (Leaf Blast Severity and Panicle Blast Incidence), ellipses are auxiliary variables, white boxes are parameters, with dotted lines indicating specific parameters for leaf and panicle blast. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$IP_{state} = \sum_{i=0}^{n} IP_{rate} \tag{1}$$

$$IP_{rate} = \begin{cases} \dfrac{1}{0.59*T_h - 8.5} \; if \; 20.83 < T_h < 35 & and \; RH \geq RH_{ex} \\ \dfrac{1}{(-0.25*T_h + 14.35)} \; if \; T_{DO_min} < T_h < 20.83 & and \; RH \geq RH_{ex} \\ 0 \; else \end{cases} \tag{2}$$

Where IP_{rate} is the hydro-thermal time (Eq. (2)), calculated as a function of hourly air temperature (T_h, °C) when air relative humidity (RH, %) is above a threshold (RH_{ex}, %) and T_{DO_min} is the minimum air temperature for disease onset (°C). When IP_{state} exceeds a user-defined threshold, leaf blast onset is triggered and the infection process starts to be simulated. For leaf blast, the threshold was calibrated using the least square method by comparing simulated average value and observed leaf blast onset dates in all experiments in three years and three locations, obtaining the average value of 150 °C h^{-1}. We assumed that inoculum was not limiting giving that rice blast disease is endemic in the area.

The environmental suitability for the infection process is simulated according to leaf wetness duration and temperature using the hourly version of the infection model developed by Magarey et al. (2005) and proposed by Bregaglio and Donatelli (2015). This model computes an air temperature response function (f(T), unitless, Eq. 6 in Bregaglio and Donatelli, 2015) modulated by wetness duration requirement (f(LW), hour, Eq. 8 in Bregaglio and Donatelli, 2015).

At each hour, a cohort of spores potentially leading to a new infection is added if the leaf is wet and $f(T) > 0$, and an infection event is completed if the wetness duration requirement is reached. The model considers the impact of a dry period on the infection process via a critical interruption value (D50, hour, Eq. 9 in Bregaglio and Donatelli, 2015).

When an infection event is completed, the latency process starts to be simulated as a function of air temperature. The duration of the latency period (LP, hour, Eq. (3)) is computed according to the same temperature response function used for infection (Eq. 6 in Bregaglio and Donatelli, 2015) with specific cardinal temperatures, considering a minimum duration of the latency period (LP_{min}, hour) at optimal temperature.

$$LP = (LP_{min}) + (LP_{min}*(1 - f(T))) \tag{3}$$

When the latency period ends, the sporulation efficiency (SP, unitless, 0–1, Eq. (4)) is computed as a function of air temperature and relative humidity (Eq. (4)).

$$SP = \begin{cases} \left(\dfrac{T_{max_LP} - T_h}{T_{max_LP} - T_{opt_LP}}\right)\left(\dfrac{T_h - T_{min_LP}}{T_{opt_LP} - T_{min_LP}}\right)^{(T_{opt_LP} - T_{min_LP})/(T_{max_LP} - T_{opt_LP})} \; if \; RH > RH_{min} \\ 0 \; else \end{cases} \tag{4}$$

Where T_{max_LP}, T_{opt_LP} and T_{min_LP} (°C) are the cardinal temperatures for sporulation.

The spore dispersal efficiency (SP_d) is then simulated considering precipitation and wind speed as driving variables. Two equations compute spore detachment by precipitation (DLrain, unitless, 0–1, Eq. 24 in Bregaglio and Donatelli, 2015) and wind (DLwind, unitless, 0–1, Eq. 26 in Bregaglio and Donatelli, 2015).

At the end of the day, the rate of hourly increase of leaf blast severity and panicle blast incidence (IR_d) is computed (Eq. (5)), using an empirical coefficient (R_v, 0–1, unitless) to simulate the susceptibility of the rice variety to blast disease and a maximum increase in leaf severity and panicle incidence (Inr, 0 - + ∞; Inr_L and Inr_P in Table 4).

$$IR_d = \begin{cases} \sum_{i=1}^{n=24} (Inr^{*}R_v^{*}\mathrm{SP}^{*}(\mathrm{DL_{wind}} + \mathrm{DL_{rain}})) \; if \text{ infection event} = \text{true} \\ 0 \; elsewhere \end{cases} \quad (5)$$

2.3. Field experiments used in model calibration and evaluation

The agrometeorological data related to the field experiments were presented in Bregaglio et al. (2017) and are summarized in Table 1, as well as the details of experimental trials used in model calibration and evaluation, i.e. sites, locations, years, and sowing dates.

Field trials were performed in 2013–2015 in the three sites of Collobiano, Confienza, and Garbagna Novarese by the Research Centre for Plant Protection and Certification of the Council for Agricultural Research and Economics (CREA-DC). Average air temperature during the rice-growing season in the three years ranged between 20 and 22 °C, whereas large variations in precipitation (206–514 mm) and average relative humidity (69–82%) were registered. Four rice varieties were grown in each site, characterized by (i) variable susceptibility level to rice blast and (ii) different crop cycle lengths (Table 2).

The level of susceptibility of the four cultivars to rice blast has been derived from Titone et al. (2015). Rice sowing was performed by manual broadcasting of soaked seed on flooded paddies. No fungicide treatments were applied. Weeds were controlled in pre-sowing (Oxadiazon, Ronstar FL, 0.75 l ha^{-1}) and post-emergence (Penoxulam, Viper, 2 l ha^{-1}). The trials were carried out under flooded conditions from sowing to the start of grain filling; then, paddies were drained to allow mechanical harvest. Experimental plots were harvested at physiological maturity, identified by 22% relative humidity of rice grains, assessed using an infrared thermogravimetric moisture meter (Sartorius MA 150) on a grounded sample of 5 g. Fertilizations were split in pre-sowing and two top dressing events, corresponding to standard farmer practices in the area, aimed at assuring no-limiting conditions of nitrogen availability to the rice crop (total dose ranging from 121 to 193 kg N ha^{-1}).

Starting from July 1 to September 30, the four top leaves (Surin et al., 1991) were weekly assessed on 20 randomly selected plants according to a standard scoring system for leaf blast severity (Lan et al., 2016) ranging from 0 to 9 (Table 3).

From the early ripening stage to harvest, an equal sample of randomly chosen plants was evaluated for quantifying panicle blast incidence (Ishihara et al., 2014), with a value ranging from 0 (no diseased plants) to 100% (100% diseased plants). Leaf blast severity and panicle blast incidence data were then averaged for each plot and sampling date, and the standard deviation was computed to obtain a single value and associated variability.

Hourly weather model inputs were estimated from daily data recorded by the over turf standard meteorological stations of the Regional Agency for Environmental Protection (ARPA, Assessorato Agricoltura – Settore Fitosanitario) placed in proximity of each experimental field. Hourly air temperature (°C) and air relative humidity (%) were estimated according to Campbell (1985) and Allen et al. (1998), respectively. Hourly leaf wetness sensors were not available; therefore, we assumed wet an hour with RH than 95% (Wolf and Verreet, 2005) or during rain.

Table 2
Rice varieties considered in model calibration and evaluation with main characteristics (vegetative and reproductive cycle length and susceptibility to leaf and panicle blast).

Variety	Vegetative cycle (days)	Reproductive cycle (days)	Leaf blast susceptibility	Panicle blast susceptibility
Vialone Nano	97	58	High	High
Deneb	80	60	High	High
Balilla	100	60	Moderate	High
Gladio	85	50	Low	Moderate

Table 3
Levels of the scoring system used to assess leaf rice blast in the experimental trials.

Level	Description	disease severity %
0	No impact	0–0.5
1	Traces of disease	0.6–0.9
2	Slight impact	1–1.5
3	Little impact	1.6–2
4	Minor impact	2.1–3.9
5	Low impact	4–10
6	Discrete impact	11–25
7	Medium impact	26–50
8	Large impact	51–75
9	Very strong impact	>75

2.4. Simulation experiment design

2.4.1. Model calibration and evaluation

The field datasets were split into calibration and evaluation subsets using two methods: hold-out (Pahikkala et al., 2009, Section 2.4.1) and bootstrap procedure (Efron, 1979). For the hold-out method, we randomly choose 44% of the datasets to be included in the calibration dataset while the others were used for evaluation. In bootstrap calibration, nine random samplings were performed using 77% of the data as training and 23% as validation sets, including all four varieties in each

Table 1
The experimental trials used in model calibration and evaluation, including sites, locations, years, sowing dates, and meteorological conditions. T_{ave} = average air temperature (°C), RH_{ave} = average air relative humidity (%), R_{cum} = cumulated rainfall (mm) in the growing season (May–October).

Site	Coordinates	Year	Sowing	T_{ave}(°C)	RH_{ave}(%)	R_{cum}(mm)
Collobiano	N45° 24′, E8° 22′	2013	May 7	21.2	77.8	461.2
		2014	May 7	20.16	81.61	329.4
		2015	May 4	21.58	72.8	249.2
Garbagna Novarese	N45° 23′, E8° 39′	2013	May 7	20.96	72	206.4
		2014	May 5	21.4	71.2	514.2
		2015	May 5	22.86	73.28	325.6
Confienza	N45° 18′, E8° 32′	2013	May 6	21.48	71.65	424.2
		2014	May 6	21.04	73.72	372.6
		2015	May 7	22.55	68.62	307.8

Table 4
Calibrated parameters values, units and literature sources used to simulate the epidemiology of *Magnaporthe oryzae* B.Couch. 1. Katsantonis et al., 2017; 2. Calvero Jr et al., 1996; 3. Kim and Kim, 1993; 4. Cardoso et al., 2008; 5. Ou, 1985; 6. Sharma and Kapoor, 2003; 7. Ahn, 1994; 8. Choi et al., 1998; 9. Degenkolbe et al., 2009; 10. Biloni and Lorenzi, 2002; 11. Ghatak et al., 2013; 12. Calvero and Teng, 1992; 13. Kato, 2001; 14. Suzuki, 1969; 15. Pooja and Abhishek, 2014; 16. Viswanath et al., 2017; 17. Rajput et al., 2017; 18. Bonman, 1992; 19. Teng, 1994.

Acronym	Parameter	Unit	value	Range	Source
RH_{ex}	Relative humidity threshold for leaf wetness	%	80.00	[80100]	2
T_{DO_min}	Minimum air temperature for disease onset	°C	12.88	[8 16]	5,6
T_{max_IF}	Maximum air temperature for infection	°C	31.58	[29 35]	3,4,16
T_{opt_IF}	Optimum air temperature for infection	°C	17.87	[17 19]	18
T_{min_IF}	Minimum air temperature for infection	°C	8.00	[7 9]	16
LW_{min}	Minimum wetness duration for infection	hour	3.00	[2 8]	2,8,15,17
LW_{max}	Maximum wetness duration for infection	hour	8.80	Default	
D50	Dry duration to interrupt a wetness period	hour	4.68	[4 6]	14
LP_{minL}	Minimum latency duration (leaf blast)	day	3.00	[2 4]	10,11,12
T_{min_LP}	Minimum air temperature for latency	°C	9.00	[9 13]	10,19
T_{opt_LP}	Optimum air temperature for latency	°C	26.00	[26 28]	10,19
T_{max_LP}	Maximum air temperature for latency	°C	33.30	[33–35]	7,9
RH_{min}	Minimum relative humidity for sporulation	%	70.00	Default	
T_{max_SP}	Maximum air temperature for sporulation	°C	33.20	[29 38]	2,13
T_{opt_SP}	Optimum air temperature for sporulation	°C	18.77	[16 30]	1,14,17
T_{min_SP}	Minimum air temperature for sporulation	°C	9.00	[9 12]	13,14
WS_{min}	Minimum wind for detachment	m s^{-1}	0.71	[0.1 3.4]	2,13
WS_{50}	Wind for 50% detachment	m s^{-1}	2.00	Model default	
WS_{max}	Maximum wind for detachment	m s^{-1}	3.00	Model default	
R50	Rain for 50% detachment	mm	1.78	Model default	
Inr_L	Maximum severity increase hourly (leaf blast)	unitless	0.017	[0.01 0.3]	17
LP_{minP}	Minimum latency duration (panicle plast)	dDay	2.30	[2 4]	10,11,12
Inr_P	Maximum incidence increase hourly (panicle blast)	%	0.24	[0.01 0.3]	17

sampled site × year combination. Automatic calibration of model parameters was performed by minimizing the Root Mean Square Error (RMSE) between simulated and observed disease data. We calibrated model parameters via gradient descent optimization algorithm using RMSE at each sampling date as the objective function. We varied the parameters according to the ranges found in the literature (Table 4) and obtained the same set for leaf and panicle blast, except for three parameters, i.e., the maximum hourly increase in severity/incidence, the minimum duration of latency and the threshold of degree hours for disease onset; the latter was calibrated using all the field datasets. For the highly susceptible variety, Vialone Nano, the susceptibility coefficient *Rv* (Eq. (3)) was set to 1. For the other varieties, the same coefficient was optimized within a [01] range. Model performances were quantified by the following metrics: mean absolute error (MAE, minimum and optimum = 0; maximum = +∞), RMSE (minimum and optimum = 0; maximum = +∞), the coefficient of residual mass (CRM, minimum = −∞, optimum = 0, maximum = 1) and Nash-Sutcliffe modelling efficiency (EF, minimum = −∞, optimum and maximum = 1, Nash and Sutcliffe, 1970), the relative RMSE (RRMSE, minimum and optimum = 0; maximum = +∞).

Further evaluation of simulation results was carried out using observations collected by CREA-DC technicians in 2008 after full flowering of crops, approximately at milk stage, during the field assessments needed for seed certification. These surveys were not carried out for modelling purposes and reported an overall assessment of panicle blast incidence close to harvest time, without the specification of the phenological stage at the time of sampling. The incidence of panicle blast was visually assessed on rice fields located in the provinces of Vercelli and Novara. The dataset included 76 rice crops consisting of 37 different varieties with varying degrees of resistance to the disease. Four seed crops were identified as Resistant, 20 as Moderately Susceptible and 52 as Susceptible, according to Titone et al. (2015). When necessary, spatialized simulations were averaged, considering the overlapping of the grid cells within the municipalities' boundaries to perform the comparison.

2.4.2. Climate change scenario assessment

After calibration and evaluation, spatial simulations with the ELLE model were performed over the whole Northern Italian rice district, where around 90% of the national production is concentrated, considering baseline conditions (2003–2014) and future climatic projections. Spatially distributed simulations were performed at 2 × 2 km resolution within a regular grid (total 1346 simulation units). The baseline weather data were extracted from the Regional Agency for Environmental Protection (ARPA) database at the same 2 × 2 km spatial resolution. Daily variables were used to estimate hourly data using the same procedure used for calibration and evaluation (Section 2.3). The baseline climate data were fed into the CLIMAK stochastic weather generator to produce 20-years synthetic climatic series referred to current and future conditions, centered on two time frames, 2030 and 2070. The monthly anomalies of temperature and precipitation corresponding to the combination of the two RCPs and four General Circulation Models (GCMs) were then used to produce future climatic weather series. The four GCMs were the Norwegian Earth System Model (NOResm, Tjiputra et al., 2013), the Model for Interdisciplinary Research on Climate (MIROC, Watanabe et al., 2011), the Hadley Centre Global Environmental Model version 2 (HadGEM, Collins et al., 2011), and the GCM developed by the Goddard Institute for Space Studies (GISS, Schmidt et al., 2006). The two extremes of the Representative Concentration Pathways (RCP) from the Intergovernmental Panel on Climate Change were considered in this study, i.e., the RCP 2.6 W m^{-2}, which corresponds to low climate change impacts and air CO_2 concentration up to 420 ppm in 2100, and the RCP 8.5 W m^{-2}, which is the worst-case scenario with air CO_2 concentration up to 936 ppm in 2100 (van Vuuren et al., 2011). The two scenarios were chosen because they account for the maximum variability in radiative increase and CO_2 concentration. Simulations were performed using a factorial combination of (i) three levels of blast susceptibility (Titone et al., 2015) – high (Vialone Nano), moderate (Balilla) and low (Gladio) - and (ii) three durations of the crop cycle (Mongiano et al., 2019) - early (825 °C day^{-1}), medium (866 °C day^{-1}), and late (983 °C day^{-1}). The blast susceptibility levels used in climate change simulations were derived from the field-level calibration and are supported by Titone et al. (2015) and Mongiano et al. (2020), who assessed the blast susceptibility on the same

varieties in dedicated experimental trials. The thermal requirements leading to different rice cycle duration were derived from Mongiano et al. (2019), who developed 13 variety-specific parameter sets for the WOFOST_GT model (Stella et al., 2014) and calibrated and evaluated them in the Italian rice-growing area. These varieties are among the most cultivated in Italian environments and represent the current Italian rice germplasm (Mongiano et al., 2018). The total number of simulated years was 3,997,620as the result of 1346 grid cells × 4 GCMs × 2 RCPs × 2 time frames × 3 susceptibility levels × 3 cycle lengths, plus baseline simulations (1346 grid cells × 3 susceptibility levels × 3 cycle lengths).

2.5. Data analysis

Data analysis was performed using the R software environment (R Core Team, 2019). Boxplots were created using the *ggplot2* R package (Wickham, 2016) to show the median (central line), the interquartile range (IQR, lower and upper hinges, corresponding to 25th and 75th percentiles), and the minimum and maximum values within 1.5 IQR from the hinges (whiskers). Data falling outside the hinges are represented as single data points and considered as outliers. GCM, RCP, Timeframe, and Phenology and Susceptible factors were ranked according to their contribution in predicting leaf and panicle blast by fitting a Random Forest model (Breiman, 2011) on simulated leaf blast severity and panicle incidence. The two models were trained using 80% of the data, leaving the remaining 20% for model evaluation. Model training was performed using the R package *party,* which allows for computation of conditional variable importance as the mean decrease in model accuracy resulting from the sequential permutation of variable predictors. Variable permutation importance is obtained via random permutation of variable X_i and calculation of the loss in prediction accuracy from a new model with the permuted variable. The rationale is that if X_i is associated with the response, its permutation leads to a decrease of prediction accuracy, then providing a measure of variable importance. The conditional procedure differs from the standard permutation principle being less biased toward highly correlated predictor variables. The tuning parameters *ntree* (number of trees to grow) and *mtry* (the number of predictors that will be randomly sampled at each split during tree models creation) were set to 50 and 3, respectively (Strobl et al., 2007a, 2007b, 2008; Breiman, 2011).

3. Results

3.1. Model performances in reproducing field data of leaf and panicle blast

The average metrics quantifying ELLE model performances are presented in Table 5.

Supplementary Table S1 presents the comparison of ELLE performances considering bootstrap resampling, which were in line with the ones obtained with hold-out method, i.e., differences of MAE below 0.3 and RMSE below 0.4 (IRRI scale) for leaf blast severity and of MAE below 4% and RMSE below 2% for panicle blast incidence. Complete information on the results obtained in each year and site is provided in Supplementary Tables S2.

The ELLE model reproduced observed leaf blast severity (IRRI scale) in the calibration datasets with RMSE values in the range 0.52–1.64 and MAE ranging between 0.29 and 1.19. RMSE values in model evaluation varied from 0.67 to 1.01, with MAE ranging from 0.36 to 0.71. RRMSE was always below 15% in both calibration and evaluation. EF values denoted accurate results, especially on the highly susceptible variety Vialone Nano (0.61 in calibration, 0.85 in evaluation). In contrast, the model generally underestimated leaf blast severity on the susceptible variety Gladio (CRM ranging between 0.55 in calibration to 0.70 in evaluation), with low EF values (−0.11–0.05). The two varieties with intermediate susceptibility obtained better results in model evaluation, with EF values above 0.5 and CRM values close to 0. The model accurately reproduced field observations on panicle blast, leading to average EF values always above 0.5, except in the Gladio calibration dataset (EF = 0.18). Average CRM denoted a slight overestimation for Vialone Nano (−0.039), Balilla (−0.26) and Gladio (−0.59). RMSE and MAE values were higher in the highly susceptible Vialone Nano (23.57% and 15.50% in calibration, 20.09% and 15.94% in evaluation), while the best results were obtained with Gladio (7.13% and 6.23% in calibration, 4.86% and 3.90% in evaluation). Moreover, the average values of MAE and RMSE were similar for Balilla (MAE = 8.93 and RMSE = 12.24) and Deneb (MAE = 12.49 and RMSE = 17.09). RRMSE for panicle blast was below 15% in all combinations, except for Gladio in the calibration dataset (16.49%).

Fig. 3 presents the simulated and observed leaf blast severity in calibration and evaluation datasets. Simulations of Vialone Nano (late variety, high blast susceptibility) led to leaf blast onset on early-August in 2013, late-July in 2014 and mid-July in 2015 in the three sites. The forecasted dates of leaf blast onset were in the range ± 1 week with respect to observations. Simulated leaf blast dynamics were consistent with field data, which were characterized by a smooth increase after disease onset, followed by a steep increase in August–September, and then a plateau in proximity of maturity, when senescence processes are dominant.

ELLE succeeded in reproducing lower leaf blast severity on Vialone Nano in 2013 with respect to the other two growing seasons, despite an overestimation in Collobiano (Fig. 3a) and Garbagna Novarese (Fig. 3g), where observed leaf blast severity ranged between 0.6 and 0.9. The onset of leaf blast occurred around mid-July on Balilla (late variety, moderate blast susceptibility) in all experiments. The observed leaf blast severity was lower on Balilla than on Vialone Nano, coherently with

Table 5
ELLE performances in reproducing leaf blast severity and panicle blast incidence in calibration (C) and evaluation (E) subsets in 2013–2015. The evaluation metrics presented are the average values across years and sites (Collobiano, Confienza, and Garbagna Novarese). They are the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Coefficient of Residual Mass (CRM), Nash-Sutcliffe modelling efficiency (EF) and relative RMSE (RRMSE).

Rice variety	Use	Leaf blast severity (0–9)					Panicle blast incidence (0–100%)				
		MAE	RMSE	CRM	EF	RRMSE	MAE	RMSE	CRM	EF	RRMSE
Vialone Nano	C	1.19	1.64	0.01	0.61	5.44	15.50	23.57	−0.04	0.64	6.44
	E	0.71	1.01	0.04	0.85	3.87	15.94	20.09	−0.04	0.70	7.97
Average		0.95	1.33	0.03	0.73	4.65	15.72	21.83	−0.04	0.67	7.20
Balilla	C	0.66	0.99	−0.43	0.16	10.00	10.51	13.47	−0.31	0.66	8.05
	E	0.56	0.76	−0.19	0.51	6.82	7.35	11.01	−0.20	0.80	8.08
Average		0.61	0.88	−0.31	0.34	8.41	8.93	12.24	−0.26	0.73	8.07
Deneb	C	0.85	1.18	−0.78	−1.61	12.18	12.81	17.05	0.11	0.66	7.58
	E	0.62	0.89	−0.16	0.78	12.16	12.16	17.13	0.23	0.76	7.64
Average		0.74	1.04	−0.47	−0.42	12.49	12.49	17.09	0.17	0.71	7.61
Gladio	C	0.29	0.52	0.55	0.05	6.23	6.23	7.13	−0.82	0.18	16.49
	E	0.36	0.67	0.70	−0.11	3.90	3.90	4.86	−0.36	0.63	12.86
Average		0.33	0.60	0.63	−0.03	5.07	5.07	6.00	−0.59	0.41	14.67

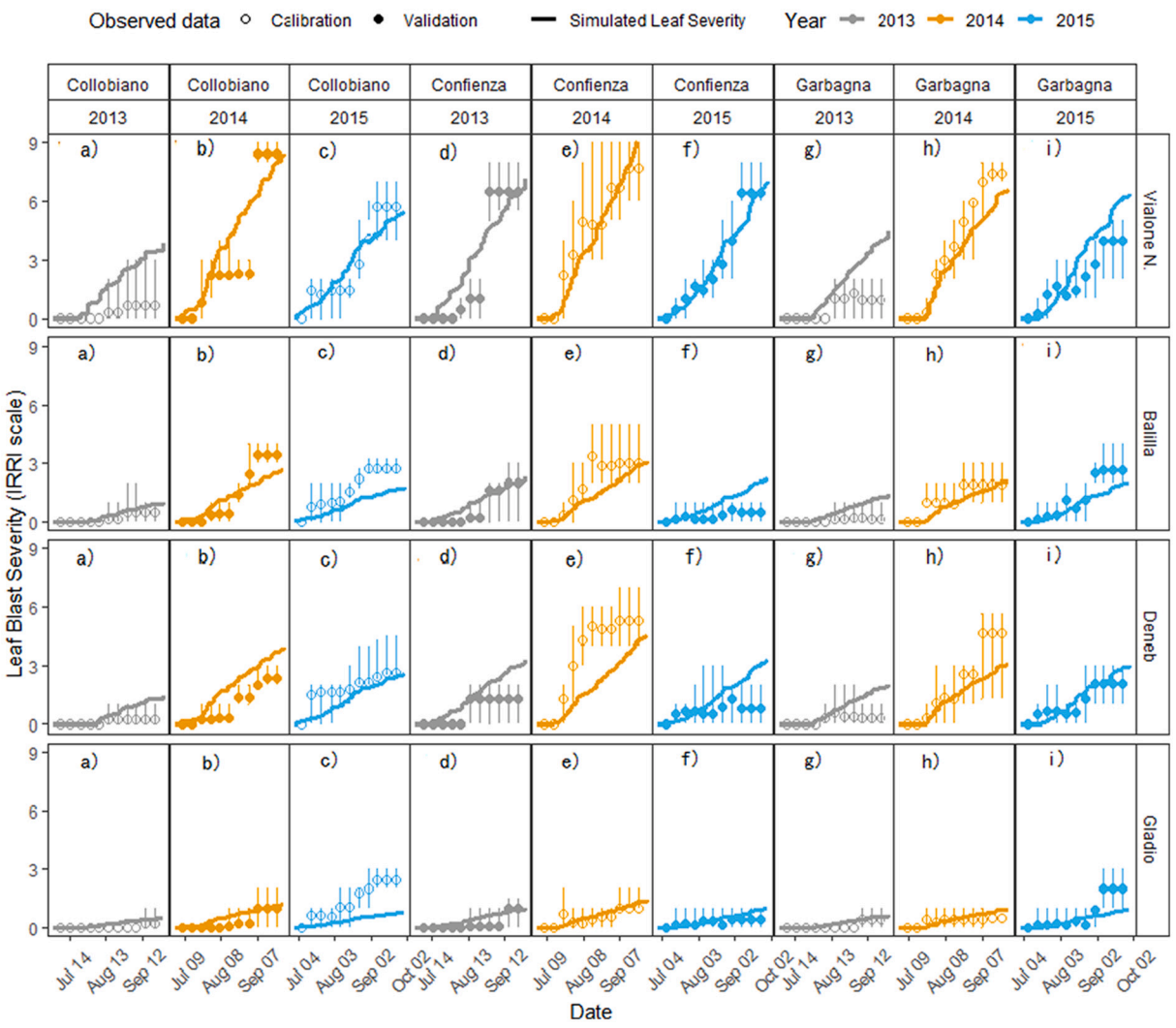


Fig. 3. Comparison of simulated and observed leaf blast severity in the three years of experimental trials. The charts along the vertical axis refer to four rice varieties (Vialone Nano, Balilla, Deneb and Gladio), the horizontal axis refer to three sites/years (Collobiano, Confienza and Garbagna Novarese).

their susceptibility level. The highest leaf blast severity was assessed in Collobiano in 2014 (Fig. 3b, 3.4), with simulations closer to field observations (Fig. 3b, 2.5). Leaf blast severity was lower in most of the other field trials, with values ranging between 2 and 3. The onset of leaf blast symptoms on Deneb (short life cycle, high blast susceptibility) occurred on mid-July in 2014 and 2015 and early August in 2013, similarly to Vialone Nano.

ELLE simulations were consistent (Fig. 3a, 1.27; Fig. 3g, 1.86; Fig. 3f, 3.32) with the very low leaf blast severity at harvest that was observed in Collobiano and Garbagna Novarese in 2013 and in Confienza in 2015 (Fig. 3a, 1; Fig. 3g, 2; Fig. 3f, 3). Leaf blast symptoms onset occurred on late July on Gladio (short growing cycle, low blast susceptibility), which was characterized by very low leaf blast severity across sites and years (scores ranging 0.2 to 1). The simulated leaf blast severity generally matched field observations, with higher prediction errors in 2015, in which the model underestimated observed data (2.4 in Collobiano, 2 in Garbagna Novarese).

Simulation of panicle blast incidence was consistent with observations in all conditions (Fig. 4), with symptoms onset appropriately simulated in mid-August. The highest incidence was observed in late-August and mid-September, especially on Vialone Nano, i.e. 100% at Confienza in 2013 (Fig. 4d), Collobiano in 2014 (Fig. 4b) and Garbagna Novarese in 2015 (Fig. 4i). The corresponding simulations matched field data, i.e. 100% (Fig. 4d), 80% (Fig. 4b) and 100% (Fig. 4i). Low panicle blast incidence was observed at Collobiano in 2013 (10%, Fig. 4a), where simulated data was 38%. High panicle blast incidence was observed in Balilla (70%, Fig. 4d; 90%, Fig. 4e) and Deneb (90%, Fig. 4d; 100%, Fig. 4e), with simulation errors around 25% (Fig. 4d and d). Panicle blast incidence in the other trials ranged between 20% and 50% for Balilla, and between 10% and 77% for Deneb. Observed temporal dynamics of panicle blast incidence on the susceptible variety Gladio were smoother, with a much lower value at harvest than in the other varieties. Minimum panicle blast incidence (5%, Fig. 4a) was recorded at Collobiano in 2013, and maximum observed values (30%, Fig. 4c) occurred in the same site in 2015.

The model evaluation at the municipality level (year 2008, Supplementary Fig. S1) proved the model ability to reproduce panicle blast incidence variability related to variety's susceptibility. The simulations with susceptible and moderately susceptible varieties closely reproduced field observations, with a slight tendency of the model to underestimate panicle blast incidence (mean difference of −4% and − 2%, respectively). On the contrary, panicle blast incidence in varieties with high susceptibility was generally overestimated with on average of 10% higher incidence, up to 25% in some cases. An extended visualization of the results of spatialized simulations performed under baseline conditions from 2003 to 2014 is provided in Supplementary Fig. S2. Unfortunately, there are no official field data for an additional evaluation, although the results agree with the official blast disease infection risk

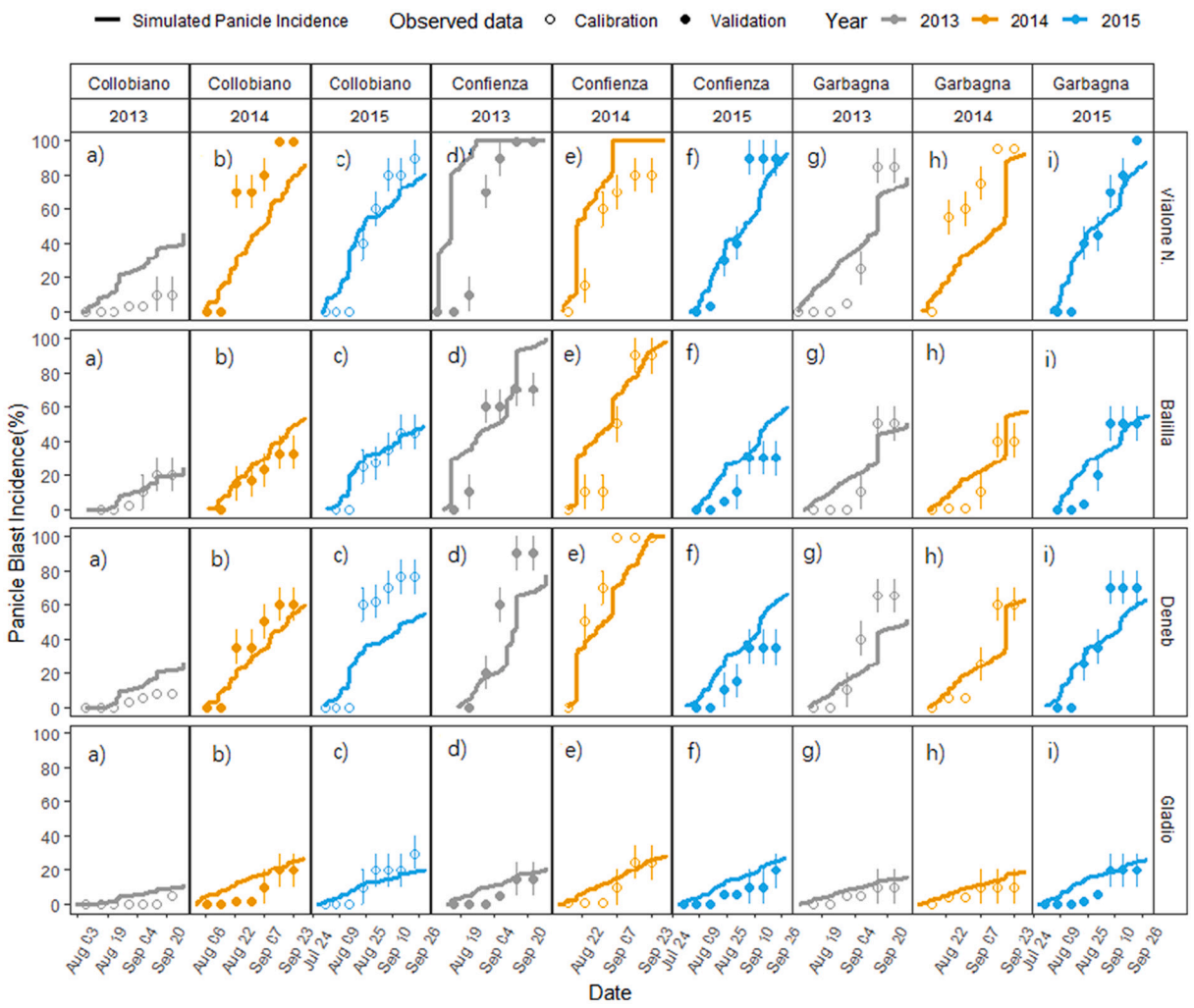


Fig. 4. Comparing simulated and observed panicle blast incidence. The charts along the vertical axis refer to four rice varieties (Vialone Nano, Balilla, Deneb and Gladio), the horizontal axis refer to three sites/years (Collobiano, Confienza and Garbgna Novarese).

bulletins issued by the competent national authorities (data not shown).

3.2. Model sensitivity to the variability of weather conditions

The sensitivity of the ELLE model to the variations in temperature and precipitation during the growing season was explored by comparing simulations outputs for intermediate crop duration, highly susceptible variety in two growing seasons characterized by contrasting weather conditions, i.e. 2003, a dry and hot season, and 2005, a wet and cold one (Fig. 5).

The 2003 growing season was characterized by high temperatures, with a north-south increasing gradient and peaks over 30 °C in some central areas of the rice district (Fig. 5a), whereas in 2005 average air temperatures were lower across the rice area and homogeneous, generally below 25 °C (Fig. 5a). In 2003, precipitations were scarce during the rice season (May–September), with minimum values less than 100 mm, except in some western areas where they reached peaks over 200 mm (Fig. 5b). In 2005 precipitations were generally higher and heterogenous, with most areas characterized by precipitations higher than 200 mm (Fig. 5b). ELLE simulations responded to the variability of weather conditions by simulating lower leaf blast severity in 2003, with values below 4 (IRRI scale) across the district and very low values (0–1) in the hottest and driest areas (Fig. 5c). Simulated leaf blast severity in 2005 was generally higher, with peaks over 6 in the district's northwest, where most precipitations and lowest average temperatures occurred (Fig. 5c). The differences between the two seasons were exacerbated considering the simulation of panicle blast incidence, which led to lower values in 2003, consistently below 40%, with a smooth West-East decreasing gradient and values below 10% in most eastern areas (Fig. 5d). In 2005, a different situation emerged from ELLE simulations, with average values above 50% on the whole district, with peaks higher than 75% in several North-Western areas. The southern part of the district, where temperatures were higher, were characterized by lower simulated panicle blast incidence, less than 25% (Fig. 5d).

In order to further investigate the sensitivity of ELLE to the variability of weather conditions, contour plots were drawn to highlight the relationships between temperature and precipitation with respect to simulated rice blast severity and incidence (Fig. 6).

High cumulated precipitation and lower temperatures led to larger severity and incidence of leaf and panicle blast. Considering the two symptomatologies, the sensitivity of the ELLE model to precipitation was larger for leaf blast severity (Fig. 6a), given that cumulated values higher than 330 mm led to values between 7 and 8 of the IRRI scale in all temperature conditions tested, from 22 to 30 °C. Similar considerations can be drawn for severity levels from 4 to 7 of the IRRI scale, where cumulated precipitation was the main factor leading to higher leaf blast severity, whereas at low cumulated precipitation (< 250 mm), higher average temperature (above 24 °C) corresponded to lower leaf blast severity (Fig. 6a). Panicle blast simulations denoted a more complex pattern concerning the combination of temperature and precipitations,

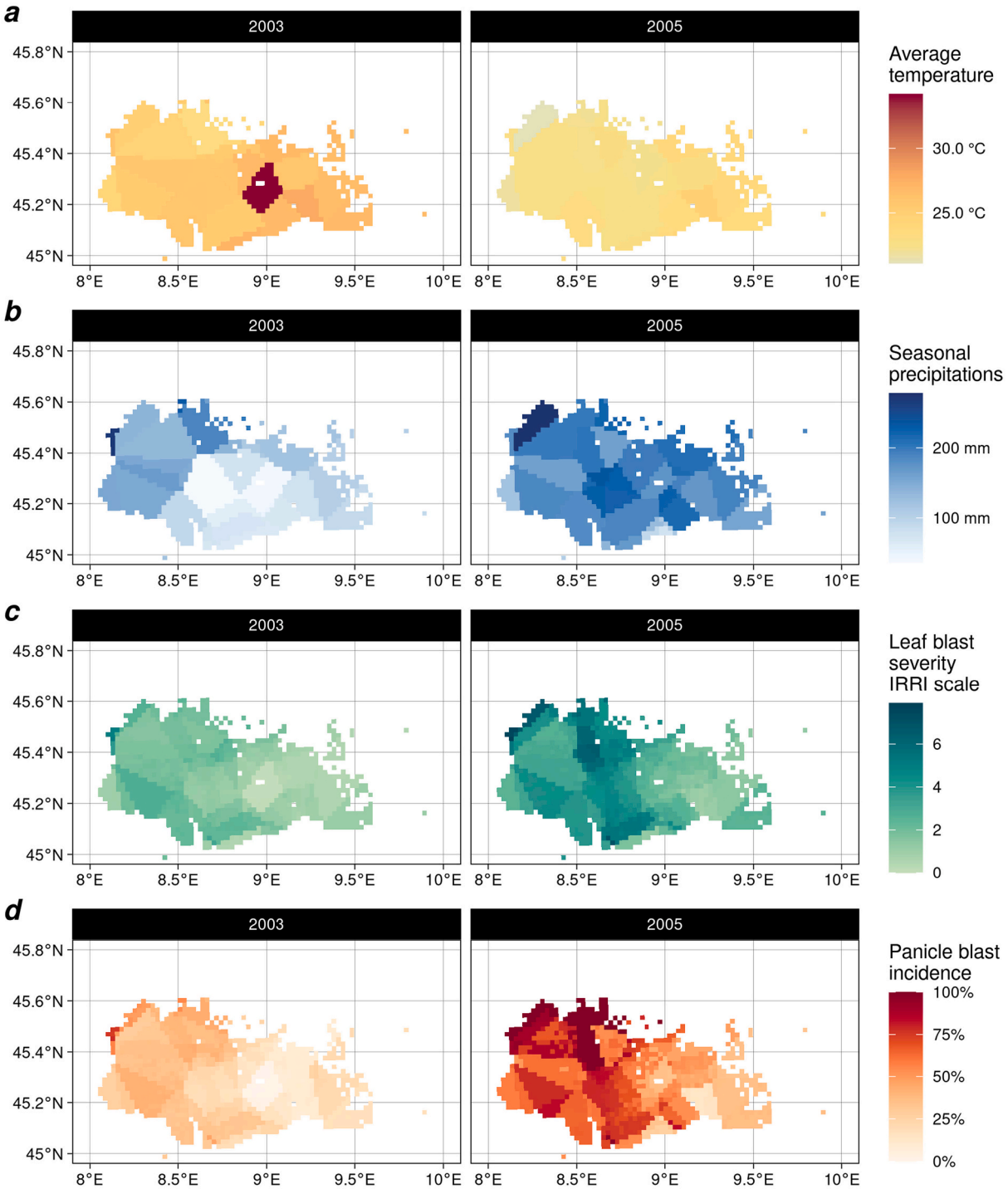


Fig. 5. Average weather conditions in 2003 and 2005 rice growing seasons (May–September) considering air temperature (a) and precipitation (b), and corresponding simulations of leaf blast severity (c) and panicle blast incidence (d).

with a simulated incidence above 90% at extreme high precipitations (> 320 mm) and low temperature (< 24 °C). Contours for the two main factors driving simulated panicle blast incidence consisted of almost diagonal isolines up to 50%. Lower panicle blast incidence (<20%) was driven mainly by high temperature (> 25.5 °C), with cumulated precipitation ranging from 100 to 250 mm (Fig. 6b).

3.3. Impact of climate change scenarios on leaf and panicle blast

The distribution of the simulation outputs from the climate change scenario analysis is reported in Fig. 7, with model results grouped by levels of varietal susceptibility, crop cycle length (Time of maturity) and time frame.

Varietal susceptibility was the largest source of variation of leaf blast severity and panicle blast incidence, with the highest impacts and related variability corresponding to the simulations where high susceptibility was considered, consistently across time frames (leaf blast severity: $\tilde{x} = 2.2$; interquartile range, IQR = 2.2; panicle blast incidence: $\tilde{x} = 32\%$; IQR = 31%). Conversely, the simulations related to susceptible varieties denoted a low impact of the disease (leaf blast severity: $\tilde{x} =$

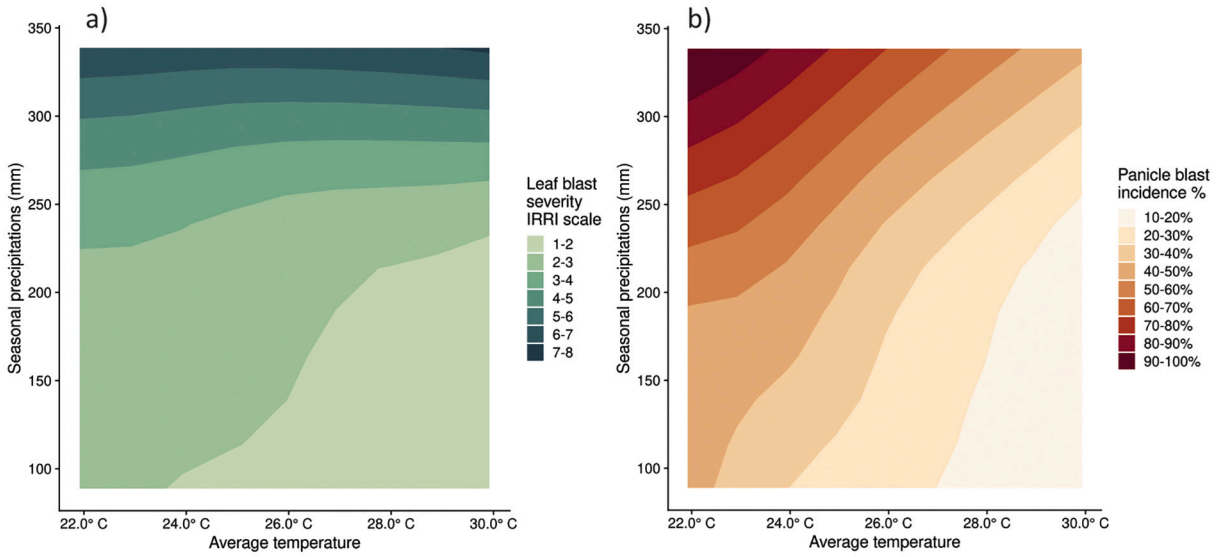


Fig. 6. Contour plots showing the relationships between average air temperature and cumulated precipitation in the rice growing season, and leaf blast severity (a) and panicle blast incidence (b). All the simulations corresponding to medium cycle varieties with high susceptibility under baseline conditions and climate change scenarios were used to draw the contour plots (12,114 simulated years).

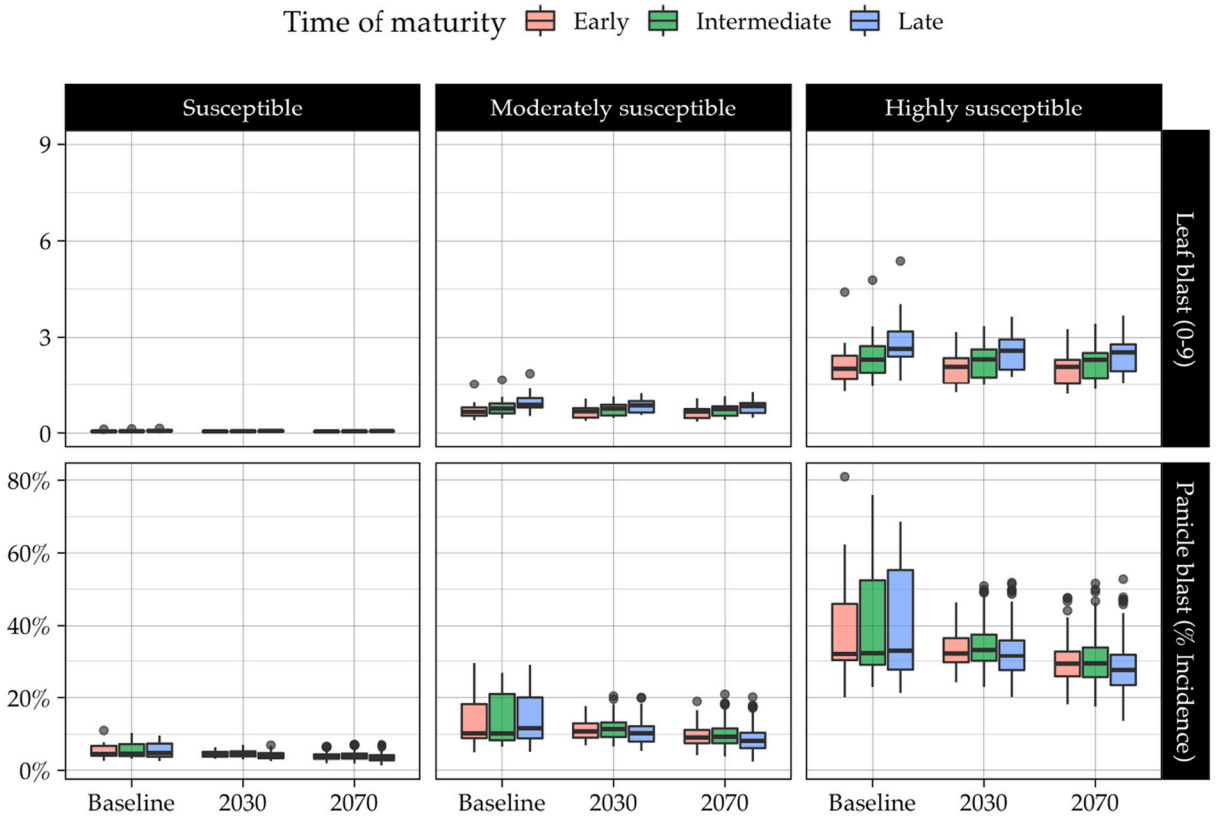


Fig. 7. Distributions of simulated leaf blast severity and panicle blast incidence grouped by timeframe and time of maturity, divided by levels of varietal susceptibility. Boxplot displays median, 25th and 75th percentiles (hinges) and 95% confidence intervals around the median (whiskers). Points correspond to values outside the 95% confidence intervals around the median calculated as (IQR•1.58)/$\sqrt{n}$.

0.05; IQR = 0.05; panicle blast incidence: $\tilde{x}$ = 4%; IQR = 4%). Early maturing varieties were associated with lower leaf blast severity, regardless of the time frame or the level of susceptibility, whereas late varieties were slightly more affected by the two symptomatologies of blast disease. The simulated leaf blast severity remained substantially unchanged in future scenarios considering all the possible combinations of GCM, RCP and time frames (Supplementary Fig. S3). A different pattern emerged for panicle blast, where small differences between the three phenology classes were simulated: the rice varieties with intermediate cycle length were the most affected by the disease (overall median incidence of 10.4%), followed by early (9.9%) and late (9.6%) varieties. Baseline conditions determined larger variability in panicle

blast incidence with larger IQR (e.g. 22.7% for highly susceptible) compared to 2030 (7.3%) and 2070 (8.5%) timeframes. The median impact of panicle blast disease remained constant in 2030, then declined slightly in the 2070-time horizon in all GCMs under RCP 8.5, but not under RCP 2.6 (Supplementary Fig. S4).

Among the considered GCMs, HADGEM and MIROC determined the most significant decline in panicle blast incidence in the 2070 time frame, whereas HADGEM simulations led to a decline of blast incidence in highly susceptible cultivars from a median value of 30.4% in 2030 to 20.5% in 2070 timeframe. To support boxplot analysis, we ranked the factors Susceptibility, Phenology, GCM, RCP, and TimeFrame by fitting a random forest model with leaf blast severity and panicle blast incidence as dependent variables. The model was very accurate in reproducing simulated data with R^2 of 93% and 94% calculated on the evaluation subset of leaf and panicle blast data, respectively. The ranking of the factors was performed considering the mean decrease in accuracy (MDA) importance score, calculated by conditional permutations of the predictors (more details in the Materials and Methods section). Results are reported in Supplementary Fig. S5 and Supplementary Table S3. Varietal susceptibility was confirmed to be the most influential predictor by two orders of magnitude for leaf blast severity (MDA = 1.7) and panicle blast incidence (MDA = 2.8%). Leaf blast severity was also predicted by the time of maturity (0.026), while GCM, RCP and time frame resulted as marginally relevant predictors in the random forest model. Considering panicle blast, both RCP (0.06%), timeframe (0.05%) and, to a lesser extent, the GCM (0.02%) scarcely influenced the prediction of simulated outputs while the effect of crop cycle length was negligible.

The results of the spatially distributed simulations concerning simulated leaf blast severity in baseline conditions and according to the climate change scenarios tested are shown in Fig. 8, considering medium cycle varieties with high susceptibility to leaf and panicle blast. The variations of weather conditions in future climatic scenarios compared to baseline climate are shown in Supplementary Figs. S6 (temperature) and S7 (precipitation).

The changes in leaf blast severity simulated in future climates ranged between −1.5 and 0.5 level of IRRI scale considering all combinations of time frame × RCP × GCM, confirming that no substantial variations are expected in the study area. The average pattern of leaf blast severity in baseline conditions showed a west-east decreasing gradient (Fig. 8a), which was reproduced in climate change simulations (Fig. 8b). Minor variations and a general slight increase in leaf blast severity have been homogenously simulated in 2030. In contrast, a larger decrease emerged in spot areas in the northern area in 2070, despite variations with

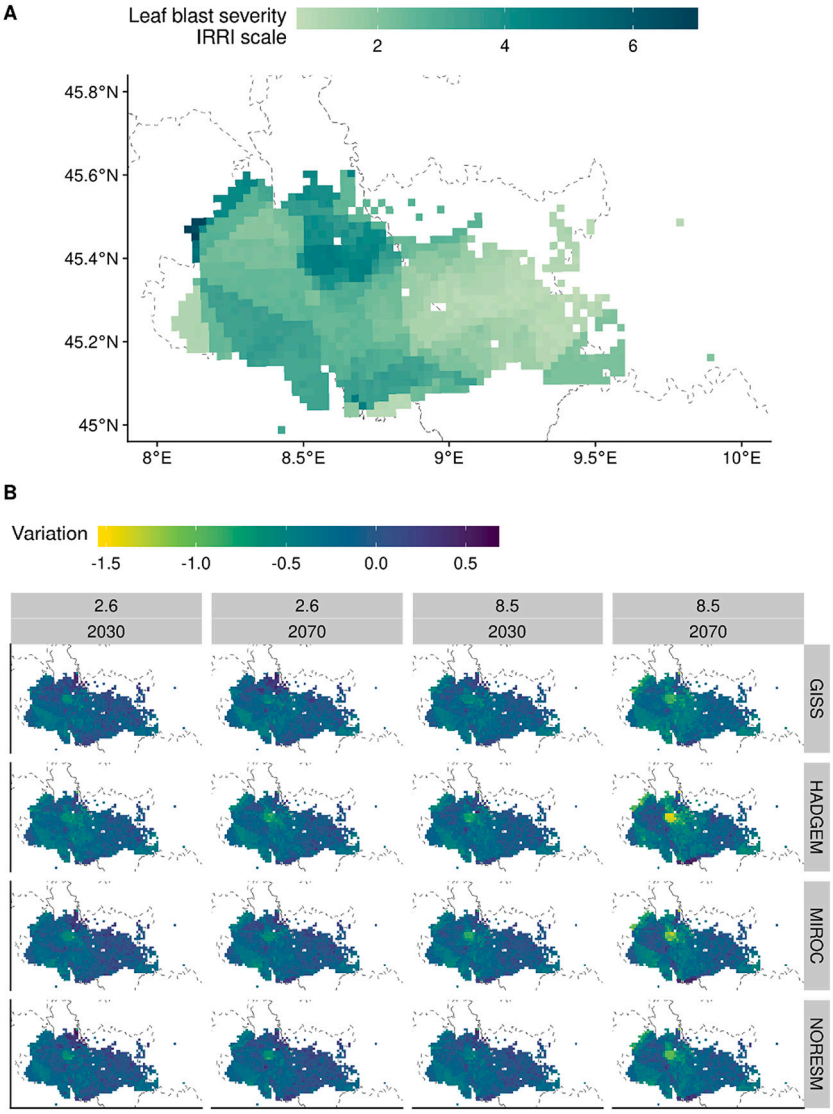


Fig. 8. Average leaf blast severity simulated in baseline conditions (A) and its variation in all the combinations of future time frame (2030, 2070), representative concentration pathways (2.6, 8.5) and general circulation models (GISS, HADGEM, MIROC, NORESM) tested (B).

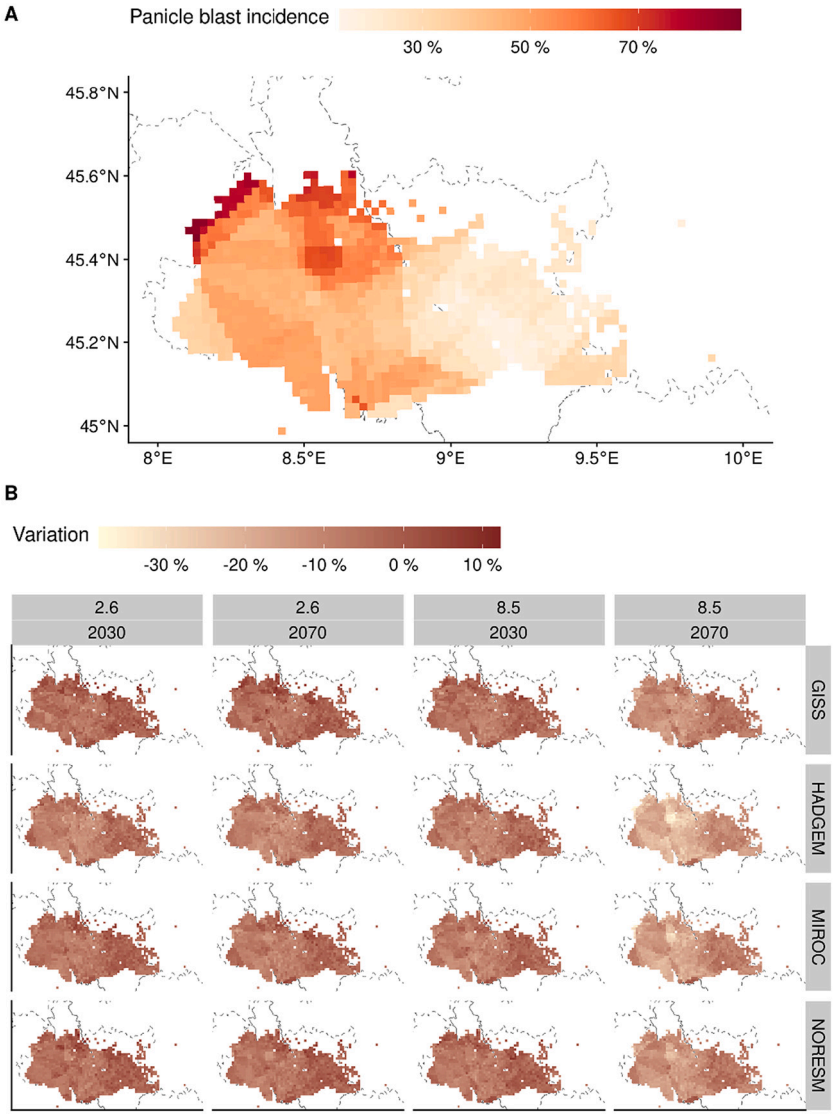


Fig. 9. Average panicle blast incidence simulated in baseline conditions (A) and its variation in all the combinations of future time frames (2030, 2070), representative concentration pathways (2.6, 8.5) and general circulation models (GISS, HADGEM, MIROC, NORESM) tested (B).

respect to baseline climate were below 1.5 level of the IRRI scale. The results of the variations of simulated panicle blast incidence in the future climate compared to baseline conditions are shown in Fig. 9.

The simulation of panicle blast incidence depicted the same west-east decreasing gradient (Fig. 9a) found for leaf blast severity (Fig. 8a). In contrast to leaf blast, simulations in climate change scenarios led to a general decrease in panicle blast incidence, especially in 2070 and considering the business-as-usual concentration pathway (RCP 8.5). HADGEM and MIROC were the GCMs corresponding to the largest decrease of panicle blast in future simulations, with peaks over −30% in the western part of the rice area (Fig. 9b).

4. Discussion

4.1. Rationale for model development

ELLE is a process-based model whose sub-models reproducing the suitability of weather conditions for epidemiological processes are parameterized according to the requirements of the rice blast pathogen. Seminal studies where the same rationale was conceived date back to the 1970s, when the first fungal models were developed, e.g., the infection rate model proposed by Kranz (1974) and the tomato/potato blight model by Waggoner and Horsfall (1969). These models share the concept that basic epidemiological processes such as the formation of spores, their dispersal and deposition by wind or rain, as well as their germination and penetration, can be simulated as a function of weather variability (e.g., temperature, relative humidity, wind speed, precipitation) via mathematic relationships. Considering rice blast, 52 studies were reviewed in 2017, the majority targeting leaf blast severity with temperature, relative humidity and rainfall as the main driving variables (Katsantonis et al., 2017). Among available models, Wang et al. (1989) developed a leaf blast model (SIMBLAST-2), composed of five sub-models, targeting rice growth, spore dispersal, latency and sporulation, using temperature, relative humidity and daylength as input of linear and nonlinear regression functions. The fungal infection process is mainly influenced by temperature and leaf wetness duration (Makowski et al., 2011; Calonnec et al., 2008), which concurrently determine the occurrence of infection. Zhao et al. (2011) developed an infection model of cucumber downy mildew by multiplying leaf wetness duration and the corresponding hourly mean temperature. Walker and Haasbroek (2007) proposed an infection model of grape downy mildew using three input variables: rainfall, air temperature and leaf wetness duration. The infection model used in ELLE is driven both by air temperature and leaf wetness duration, considering the effect of a dry period as interrupting

the infection process, and has already been used to simulate leaf blast infections (Bregaglio et al., 2016). The latency period is the asymptomatic phase between infection and sporulation (Pariaud et al., 2009), and its duration is crucial to take timely action to control the disease. It has been known for decades that the primary determinant of the length of the latent period is the temperature (Tollenaar and Houston, 1967; Johnson, 1980), with leaf wetness duration acting as a secondary factor (El Jarroudi et al., 2009). Thus, the available modelling approaches to estimate the length of the latency period are a function of the leaf or air temperature (Pfender, 2001; Yang et al., 2007; Rossi et al., 1997). Our model considers a minimum duration of the latency period at optimal temperature, which is extended when temperatures shift toward minimum and maximum cardinal values. The sporulation process corresponds to the production of new inoculum, which could lead to secondary infections, and it is mainly affected by temperature and relative humidity (Leandro et al., 2001; Sun et al., 2016). Accordingly, the ELLE sporulation sub-model considers the combined effect of temperature and air relative humidity on sporulation (Seebold et al., 2001).

The dispersal of spores is fundamental to epidemic development, and the main routes of fungal pathogens dispersal are wind-borne or splash-borne spores (Mc Cartney and Fitt, 1998). Geagea et al. (2000) explored the effect of different rainfall and rain duration on the dispersal of wheat rust. Aylor (1982) proposed a dispersal function of daily wind speed for a barley crop. Waggoner and Horsfall (1969) described a dispersal function of rainfall and leaf area index. These two models were parameterized according to rice blast epidemiology and used in ELLE, considering their compound contribution to increasing leaf blast severity and panicle blast incidence.

The choice to develop a new rice blast model rather than using existing modelling approaches is supported by two main reasons. The first is that, according to the authors' knowledge, the joint simulation of the dynamics of leaf and panicle blast has been hardly realized, mostly because in most rice-growing areas leaf blast is the main symptomatology of the disease. In Italian conditions, however, both symptomatologies are present, and panicle blast is of uttermost importance. The second reason is that the accessibility of existing models and the possibility of using them in massive simulation jobs is limited to the groups of researchers who developed them. An exception is the EPIRICE model (Savary et al., 2012), which is a Susceptible-Exposed-Infected-Removed (SEIR) model similar to the one developed in Bregaglio and Donatelli (2015), which was used as the basis for ELLE development. However, these two models suffer from the same limitations outlined above, i.e., they simulate disease severity without differentiating between leaf and panicle blast. These were the reasons why we decided to adapt existing sub-models for the epidemiological processes of rice blast to give a model capable of differentiating among the two symptomatologies of rice blast disease. Another study from the same authors focused on integrating a rice simulator with a blast disease model, where two coupling points to reproduce the damage mechanisms due to leaf and panicle blast were realized (Bregaglio et al., 2016). That study was carried out in the same area (Northern Italy), and used an SEIR model to reproduce the dynamic of disease severity without differentiating leaf and panicle blast. Given the availability of an experimental dataset where the disease severity and incidence were measured weekly, we decided here to develop a new modelling solution where the dynamics of the two symptomatologies of the disease could be distinctly simulated.

We also preferred to focus on reproducing disease severity and incidence without considering the interactions with the host crop, to avoid adding a layer of complexity without testing with reference data, given that leaf area index and aboveground biomass were not sampled at the same time resolution. Further developments of this study could envisage coupling ELLE with a rice crop model, following an experimental protocol where both crop growth and disease are frequently monitored and sampled during the season to target the physiological interactions occurring in the pathosystem.

4.2. Model parameterization and evaluation

There is no standard reference methodological procedure to be followed in the calibration of agricultural models (Seidel et al., 2018). The typical procedure consists of estimating the values of model parameters to give the best fit to observations data (Wallach, 2011) and then validate/evaluate model performances on independent datasets (Choruma et al., 2019). Here we tested two alternative way of performing model calibration, i.e., hold-out and bootstrap resampling, which gave similar results. We then decided to keep the results of the hold-out method, as it is widely used in agroecological modelling studies (e.g., Ceglar et al., 2019; Moretti de Souza et al., 2020). Given the multiple non-linearities characterizing process-based models, the parameter values must be meaningful from the ecological and biological perspective (Wang et al., 2013). We performed literature research to collect plausible ranges for the parameters driving the pathogen's processes simulated by ELLE (Table 4). Here we discuss the values of the optimized parameters in light of the ones measured or estimated in other studies. Relative humidity threshold for leaf wetness in rice crop was set to 80%, the number of days with relative humidity is over, or equal 80% was proved has highly correlated with rice blast by Calvero Jr et al. (1996) in Philippines and Hsieh (2019) in Taiwan. Minimum air temperature for disease onset was optimized to 12.9 °C and the range of variation of this parameter was set to 8–16 °C according to Sharma and Kapoor (2003). The optimized cardinal minimum (8 °C), optimum (17.9 °C) and maximum (31.6 °C) air temperature for infection were in the range reported by several authors (Rajput et al., 2017; Cardoso et al., 2008; Viswanath et al., 2017). The optimized minimum wetness duration for infection (3 h) was in the range of 2 to 8 h, according to Choi et al. (1998). Wetness duration D50 (5 h) was similar to the value (4 h) used by Suzuki (1969). Minimum latency duration was considered as specific for leaf and panicle blast and set to 3 days for leaf blast and 2.3 days for panicle blast, respectively. These values are similar to the observed values by artificial inoculation in Ghatak et al. (2013). The minimum, optimum and maximum air temperatures for the latency period were optimized to 9 °C, 26 °C and 33.3 °C, respectively. These values agreed with Degenkolbe et al. (2009), who analyzed the correlation of temperature and rice blast severity. The optimized minimum (9 °C), optimum (18.8 °C) and maximum (33.2 °C) air temperature for sporulation were in the range of available literature data: for example, a range of minimum air temperature for sporulation from 9 °C to 12 °C was observed in the lab by Suzuki (1969) and Kato (2001). The ranges of optimum (16–26 °C) and maximum (29–38 °C) air temperature for sporulation obtained by Rosangela and Ghini (2014) and Rajput et al. (2017) confirmed our calibrated values. The minimum wind speed for spore detachment (0.7 m s^{-1}) was in the same range of [0.1 3.4] proposed by Calvero Jr et al. (1996).

The ELLE model accurately reproduced leaf blast severity and panicle blast incidence in 36 experimental trials, with similar performance in calibration and evaluation subsets. When compared to field observations at the municipality level, simulation outputs could reproduce the spatial variability and account for the level of varietal susceptibility, even within a relatively small geographic area. Larger differences between model estimates and observations emerged in the rice varieties with high susceptibility; however, some factors which were not considered in simulations could have played a significant role in determining these discrepancies. Firstly, the assessments performed by CREA-DC technicians reported an overall judgment of disease severity close to harvest without specifying the phenological stage at the time of the sampling, making impossible the exact comparison of observations with simulation results. Also, agronomic management was not considered in our study, whereas one or more fungicide sprays are typically applied on susceptible cultivars. These factors could explain the lower panicle blast incidence in the observations compared to simulated data, despite the overall model agreement.

There were cases of under- and over-estimation of rice blast

observations in the field level application of the ELLE model (Figs. 3 and 4). The main reason is that the automatic calibration performed to optimize ELLE parameters found a compromise between the global model error across datasets, leading to a single set of values to reproduce pathogen biological requirements and varietal susceptibility. This means that each variety has been characterized by a single susceptibility coefficient (*Rv*, Eq. (5)), which has been used to modulate the increase in rice blast severity/incidence. Vialone Nano emerged as the most susceptible variety ($Rv = 1$), and Gladio was the less susceptible ($Rv = 0.44$). When observations on Vialone Nano were close to the other varieties, the ELLE model generally overestimated field data. This happened, e.g. in 2013, when observed leaf blast severity (< 1 IRRI scale) and panicle blast incidence (10%) on Vialone Nano were very low in Collobiano, despite with high variability (Fig. 3a). Corresponding simulations of leaf blast severity yielded a model overestimation of about 2 in the IRRI scale, which correspond to less than 2% of diseased leaf tissue (Table 2). Simulated panicle blast incidence was also underestimated by about 20%, despite Collobiano 2013 being the conditions in which ELLE simulated the lowest blast impacts. Another possible source of errors in our simulations is represented by the lack of explicit consideration of nitrogen availability to the crop, which could have modulated leaf blast severity and panicle blast incidence, despite the agricultural management of experimental field trials aimed at assuring no nitrogen stress. The correlation between observed and simulated leaf severity and panicle blast incidence was always highly significant ($p < 0.001$), with Pearson's r coefficient ranging from 0.54 (Gladio) to 0.88 (Vialone Nano) for leaf blast, and from 0.6 (Gladio) to 0.87 (Balilla) for panicle blast (Supplementary Fig. S8).

4.3. Main factors driving leaf and panicle blast impacts in climate change scenarios

The simulated future rice blast epidemics underlined the pivotal role of varietal susceptibility in modulating rice blast epidemics. In contrast, the expected trends in the short-term future were almost unchanged, and slightly declined in 2070 time horizon considering panicle blast incidence. These results agree with Kim et al. (2015), who concluded that varietal susceptibility primarily influences rice blast impacts, although it had limited integration in forecasting models so far (Katsantonis et al., 2017). Genetic tolerance to blast disease has been the main driver of Italian rice breeding throughout the twentieth century and until today (Faivre-Rampant et al., 2011; Mantegazza et al., 2008). However, most cultivated rice varieties nowadays still present low to moderate resistance to the disease (Titone et al., 2015; Mongiano et al., 2020). In our simulations, varietal susceptibility outweighed the differences due to the expected future trends under climate change scenarios. Our results agree with literature evidence, which also reports that genetic resistance is the most effective method of managing rice blast to date (Deng et al., 2017; Zhao et al., 2018). We reproduced the different level of varietal susceptibility to rice blast through an empirical coefficient regulating the increase in severity/incidence, which was calibrated to reproduce field experimental data. However, the ELLE model can be specialized by including distinct coefficients to modulate the varietal susceptibility to leaf and panicle blast, given the evidence that differences exist between the expression of resistance according to the plant organs (Ghatak et al., 2013). Analyses of isolates from infected rice plants found a large variability of races concurrently present in the Italian rice fields, with different temperature response, morphology, and pathogenicity (Crispino et al., 2012). This aspect would even exacerbate the challenges that rice breeders have to face in the coming future, also in light of the capability of the pathogen to develop fungicide resistance and periodically overcome genetic resistance (Titone et al., 2015; Zhang et al., 2006). Indeed, we did not implement the evolutionary potential of the blast pathogen in ELLE, which could be capable of rapidly adapting to changes in climatic and management conditions.

The few available simulation studies assessing future rice blast impacts indicate that the disease risk is projected to either increase or decrease depending on the geographic region considered due to the favorability of climatic conditions for the blast pathogen. In the areas where climatic conditions are already suitable for rice blast disease, the projected impacts of climate change are expected to reduce the disease risk due to increased air temperatures above the optimal ranges for the pathogen. Kim et al. (2015) used the EPIRICE model calibrated in South Korea in a climate change scenario assessment, revealing a decreasing trend in leaf blast risk due to the thermal increase projected in the 21st century, thus highlighting an inverse relationship between air temperature and leaf blast risk. Conversely, in the areas where climatic changes will be associated with a more favorable thermal regime for the pathogen's epidemiological processes, then the disease impact is expected to increase (Luo et al., 1998; Viswanath et al., 2017), at the same time leading to earlier host infection and colonization (Caubel et al., 2017). Another relevant factor influencing plant disease severity is the leaf area index, whose dynamic was not considered in our simulations.

Given that leaf blast symptoms occurred in the field experiments close to the flowering stage and that we did not have the facilities to sample leaf area index dynamically, we decided not to consider its dynamics in this study explicitly. However, coupling ELLE with a crop simulation model could provide a daily value of leaf area index, which could then be used to modulate the disease severity according to host growth, contributing to the assessment of yield losses as done in the models WHEATPEST (Willocquet et al., 2008) and RICEPEST (Willocquet et al., 2002). In addition, the climate change scenario analysis was performed testing different crop cycle duration as from Mongiano et al. (2019), who calibrated the WOFOST_GT crop growth model to reproduce the phenology and growth of 13 cultivars, representative of the Italian rice landscape. Given that the phenology module of WOFOST_GT did not explicitly simulate the occurrence heading stage, we did not include heading precocity as a source of variability in our simulation experiment, despite evidence of severe damages to the panicles after infections occurred during the heading phase (Hayashi et al., 2019).

Our study highlights a strong relationship between simulated rice blast epidemics with temperature and precipitation. Precipitation increase has been positively correlated with leaf blast severity and panicle blast incidence, with values over 200 mm corresponding to high disease impact. Low temperatures were associated with higher disease severity, with panicle blast as the most sensitive symptomatology.

The future distribution of temperature and precipitation in climate change scenarios projected a similar overall impact of leaf blast severity, highlighting that inter-annual variability of the impact of the disease will remain considerable. In an environment where the current climatic conditions are over-optimal for rice blast, the projected temperature increase in 2070 marginally affected the variability of leaf blast severity. As far as we know, no studies focused yet on the effects of climatic changes on panicle blast, which is the symptomatology most impacting rice yield in the tested conditions. The occurrence of panicle blast is widespread in the Italian temperate climate as it is favored by temperature drops at night and high relative humidity that often occur during rice grain filling.

A detailed analysis of the expected changes in the model outputs at the level of the simulated epidemiological processes is not provided in our study. This could be the main topic of further work, where reference data of infection, sporulation efficiency, and spore dispersal are used to evaluate model responses.

4.4. Using the BioMA framework to develop disease simulation models

This study extends the *Diseases* software components presented in Bregaglio and Donatelli (2015), which are developed using the Biophysical Models Application framework (BioMA, Donatelli et al., 2012). According to the terminology used in the seminal paper, we realized the ELLE model as a pathosystem-specific *composite strategy*, by associating the *simple strategies* (i.e., models, Table 1 in Bregaglio and Donatelli,

2015) related to symptoms onset (*DiseaseOnset*), infection (*InfectionWD*), latency (*Latency*), sporulation (*SporulationRH*) and spore dispersal (*DispersalWind*, *DispersalRain*).

The ELLE model is released as RESTful Application Programming Interfaces (APIs) that can be used in operational public services and accessed by third parties' private players upon request to the corresponding author. The pillar of this IT infrastructure is the modularity of the various software components, which can be either used in standalone applications or coupled in a single modelling solution, as in the ELLE model, and run against different weather data sources without modifying their core. The BioMA models are being released as "Software as a Service" (SaaS), using the Microsoft Azure Functions to allow parallel cloud computing within the Microsoft Azure suite. The response to a single call to Azure Functions is processed in Microsoft's Azure cloud, and its call and response take the form of an *HTTP* request, available after obtaining a personal call token (https://developer.progettoagridigit.it/).

5. Conclusions

Our study evaluates the current and future trends of leaf and panicle blast in the main Italian rice district by applying a new process-based simulation model. The model evaluation against field observations denoted accurate reproduction of leaf blast severity and panicle blast incidence considering rice varieties with different combinations of crop duration and disease susceptibility. The model application under an ensemble of climate change scenarios highlighted that (i) varietal susceptibility will remain the most influential factor in modulating future trends of leaf blast severity and panicle blast incidence, and (ii) disease impacts are expected to remain similar to current conditions in Italian rice-growing areas. These results emphasize the need to invest resources on plant breeding programs to release rice varieties with improved blast resistance while confirming the relevance of the disease in influencing rice cultivation in the next future.

Declaration of Competing Interest

The authors of this paper certify that they have no affiliations with or involvement in any organization or entity with any financial interest in the subject matter or materials discussed in this manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.agsy.2021.103223.

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